

Task 1 – STRATEGIC PLANNING AND DATA EXPLORATION IN LOGISTICS

Program: Yuva Intern

Internship Area: Logistics Data Analysis

Case Study: Indian Rail Freight Movement

Dataset: Inter/Intra-State Movements/Flows of Goods by Rail, River and Air – 2024-25

Primary Source: Directorate General of Commercial Intelligence & Statistics (DGCIS), Government of India

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1. Introduction

Logistics is not only about moving goods from one place to another. It also involves deciding where capacity is needed, which routes are carrying the largest loads, which commodities are driving demand, and where future pressure on the network may appear. Data can make these decisions more systematic.

For this task, I selected an Indian government dataset on the movement of goods by rail, river and air and will concentrate on the rail component. The choice is intentional because rail freight provides a clear example of a large-scale logistics network with measurable origin, destination, commodity and quantity patterns. The analysis proposed in this report is designed as the planning stage of a larger Python-based logistics analytics project.

2. Background Research

The Directorate General of Commercial Intelligence & Statistics (DGCIS), under the Ministry of Commerce and Industry, publishes annual statistics on inter-state and intra-state movements of goods by rail, river and air. DGCIS states that the publication is compiled annually on a financial-year basis and that the rail statistics are based on returns received from railway authorities. The publication covers selected commodities and presents rail movement both as state-wise inward/outward information and as origin-destination flow tables.

The 2024-25 publication reports 15,749.90 lakh tonnes of goods transported by rail within India, including internal movement. This was 2.06% higher than the 15,431.91 lakh tonnes reported for 2023-24. The report also shows that a relatively small number of commodity groups account for a large share of rail freight, which makes commodity concentration an important area for further analysis.

This project therefore treats the DGCIS publication as a starting point rather than as a complete operational dataset. The report provides useful aggregate freight statistics, but variables such as real-time train position, transit time, fuel cost, terminal congestion and wagon availability are outside the scope of the public data used here.

3. Project Definition

3.1 Proposed Logistics Scenario

The proposed scenario is a logistics planning team responsible for understanding India's rail freight movement. The team wants to answer three practical questions: where is freight originating, where is it going, and what types of goods are creating most of the demand?

The answers can support later decisions about capacity planning, freight terminals, resource allocation and identification of important freight corridors. At this stage, the project will not attempt to change railway operations directly. Instead, it will build an analytical framework that can be extended when more operational data becomes available.

- Compare total, inward, outward and internal rail freight across states.
- Identify commodities that contribute most to inter-state and internal movement.
- Identify major origin-destination freight flows.
- Group states with similar freight profiles using clustering.
- Use historical editions to investigate possible freight-demand trends.

- Translate the analytical results into practical logistics questions.

4. Dataset and Source

The selected source is the DGCIS publication titled “Inter/Intra-State Movements/Flows of Goods by Rail, River and Air – 2024-25.” The full publication is available through the DGCIS website, along with detailed statistical tables. The project will use the rail tables as the primary analytical input.

Data component	What it contains	Planned use
State-wise total rail movement	Total goods movement by state	State ranking and KPI calculation
Inward / outward / internal movement	Direction of freight movement	State freight profiles
Principal commodities	Movement by selected commodity groups	Commodity concentration analysis
ITC-HS chapters	Commodity grouping at chapter level	Broader commodity segmentation
Inter-state movement	Origin-destination freight flows	Corridor and flow analysis

DGCIS describes its rail-borne trade statistics as a 37×37 origin-destination representation across trade blocks, together with state-wise inward and outward consignments for selected commodities. This structure is particularly useful for building an origin-destination matrix in Python.

5. Project Objectives

1. Understand the structure of Indian rail freight movement using official 2024-25 statistics.
2. Create a clear KPI framework for monitoring freight movement.
3. Explore state-wise and commodity-wise differences using Python.
4. Identify major origin-destination flows and freight concentration.
5. Propose suitable data-science methods for segmentation and forecasting.
6. Design an analytical roadmap that can be extended into a complete logistics project.
7. Convert the analysis into findings that can support future logistics planning.

6. Key Performance Indicators

KPI	Definition	Why it is useful	Interpretation
Total Rail Freight	Total tonnes transported	Shows overall freight activity	Higher movement means higher network demand
Outward Freight	Goods sent from a state to other states	Finds major origin states	High value indicates a major source of freight
Inward Freight	Goods received from other states	Finds major destination states	High value indicates strong receiving demand
Internal Freight	Goods moved within a	Measures intra-state	Shows importance of

	state	activity	internal movement
Inter-State Share	Inter-state freight as a share of total movement	Measures cross-state dependence	Useful for corridor analysis
Commodity Share	Commodity freight divided by total freight	Shows concentration	High share means greater dependence on that commodity
Year-on-Year Growth	Change compared with previous year	Tracks changing demand	Positive/negative growth identifies trend changes

7. Initial Findings from 2024-25

The official 2024-25 DGCIS report provides several useful starting observations for the proposed analysis.

- Total rail goods movement within India, including internal movement, was 15,749.90 lakh tonnes in 2024-25.
- The corresponding figure for 2023-24 was 15,431.91 lakh tonnes, giving a reported growth of 2.06%.
- Coal, Coke and Briquettes etc., Cement/Clinker/Asbestos Cement, and Iron Ore together accounted for 67.39% of total inter-state rail movement.
- For internal rail movement, Coal/Coke/Briquettes etc., Iron Ore, and Cement/Clinker/Asbestos Cement together accounted for 79.08%.
- Jharkhand, Odisha and Chhattisgarh were the three largest origin states for outward inter-state rail movement, with reported shares of 18.57%, 17.81% and 13.25%, respectively.

These figures are descriptive findings from the government publication. They should not be interpreted as proof that a particular state or commodity is more efficient or more important in an overall economic sense. They simply show where the reported rail freight movement was concentrated during the financial year.

8. Data Exploration Plan

8.1 Data Collection

The first step will be to obtain the 2024-25 detailed statistical tables from DGCIS. If the required tables are supplied in Excel or another machine-readable format, they can be loaded directly into Pandas. If a table is available only in PDF form, the relevant table will first be extracted and checked against the published totals.

8.2 Data Cleaning

- Standardize state and commodity names.
- Convert quantities into a consistent unit, preferably tonnes.
- Check for duplicate rows after extraction.
- Identify blank, zero and unavailable values separately.

- Validate state totals against the published totals.
- Keep the original source table unchanged and perform cleaning on a separate working copy.

8.3 Exploratory Data Analysis

- Rank states by total freight.
- Compare inward, outward and internal movement.
- Calculate commodity shares.
- Identify the largest origin-destination flows.
- Create a heatmap of the origin-destination matrix.
- Compare the freight composition of states.
- Look for highly concentrated commodities and unusual state-level patterns.

9. Data Science Approach

Descriptive analytics: Establish the basic freight profile and KPI values before applying advanced models.

Exploratory data analysis: Use charts and statistical summaries to identify patterns, concentration and outliers.

Clustering: Group states with similar freight characteristics, such as high outward freight or similar commodity mixes.

Regression / forecasting: Use historical DGCIS editions to estimate future freight demand where a sufficient time series is available.

Optimization: Formulate a later-stage capacity allocation problem using freight demand and operational constraints.

10. Strategic Roadmap

Stage	Activity	Output
Business understanding	Define the decision problem and agree on KPIs.	Validated analytical input / decision insight
Source identification	Use DGCIS as the primary official data source.	Validated analytical input / decision insight
Data acquisition	Download the 2024-25 publication and detailed tables.	Validated analytical input / decision insight
Data preparation	Extract, clean and validate the tables.	Validated analytical input / decision insight
Feature engineering	Create freight shares, ratios, growth rates and flow indicators.	Validated analytical input / decision insight
EDA	Study state, commodity and origin-destination patterns.	Validated analytical input / decision insight
Segmentation	Cluster states according to	Validated analytical input /

	freight characteristics.	decision insight
Forecasting	Use multiple years where available to study demand trends.	Validated analytical input / decision insight
Optimization	Build a conceptual model for allocating capacity to high-demand corridors.	Validated analytical input / decision insight
Reporting	Convert results into charts, findings and logistics recommendations.	Validated analytical input / decision insight

11. Python Code Illustration

The following is a proposed starting script. The exact column names will be adjusted after the detailed DGCIS table is downloaded and inspected.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load the cleaned DGCIS rail table
rail = pd.read_excel("DGCIS_Rail_2024_25.xlsx")

# Standardize column names
rail.columns = (
    rail.columns.astype(str)
    .str.strip()
    .str.lower()
    .str.replace(" ", "_")
)

# Basic checks
print(rail.shape)
print(rail.head())
print(rail.isna().sum())

# State-wise total freight
state_freight = (
    rail.groupby("state", as_index=False)["quantity_tonnes"]
    .sum()
    .sort_values("quantity_tonnes", ascending=False)
)

# Freight share
total = state_freight["quantity_tonnes"].sum()
state_freight["share_pct"] = (
    state_freight["quantity_tonnes"] / total
) * 100

print(state_freight.head(10))

# Bar chart
plt.figure(figsize=(10, 6))
plt.bar(
    state_freight.head(10)["state"],
    state_freight.head(10)["quantity_tonnes"]
)
plt.title("Top States by Rail Freight Movement")
plt.xlabel("State")
```

```
plt.ylabel("Freight (tonnes)")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# Origin-destination matrix
od = pd.pivot_table(
    rail,
    values="quantity_tonnes",
    index="origin_state",
    columns="destination_state",
    aggfunc="sum",
    fill_value=0
)

plt.figure(figsize=(12, 8))
sns.heatmap(od)
plt.title("Inter-State Rail Freight Flow Matrix")
plt.xlabel("Destination")
plt.ylabel("Origin")
plt.tight_layout()
plt.show()

# Example clustering
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans

features = state_freight[["quantity_tonnes", "share_pct"]]
X = StandardScaler().fit_transform(features)

model = KMeans(n_clusters=4, random_state=42, n_init=10)
state_freight["cluster"] = model.fit_predict(X)

print(state_freight[["state", "cluster"]].head())
```

The code is an illustration of the planned workflow, not a claim that the final analysis has already been completed. The next stage of the project would be to inspect the detailed DGCIS tables and adapt the field names and calculations to the actual data structure.

12. Predictive Modeling Plan

Predictive analysis will only be introduced after the descriptive analysis is stable. For example, if multiple years of comparable DGCIS tables are assembled, annual freight movement can be modeled as a time series. Linear regression can provide a simple baseline, while tree-based regression or time-series models can be tested as more flexible alternatives.

For a forecasting problem, the evaluation should use a time-based split rather than randomly mixing past and future observations. Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) can be used to measure forecast error. The final model should be selected based on both numerical performance and interpretability.

13. Clustering Plan

Clustering can be used to identify groups of states with similar freight profiles. Possible features include total freight, inward freight, outward freight, internal freight, inter-state share and the share of major commodities. Before clustering, features should be scaled because freight quantities can differ

substantially across states.

K-Means can be used as a baseline method. The number of clusters should not be chosen only because it produces a visually appealing chart. Silhouette score and the practical meaning of each group should both be considered.

14. Optimization Concept

A later phase can convert the analysis into a resource-allocation problem. For example, suppose a planner has limited freight capacity and wants to allocate capacity among several high-demand corridors. The model could minimize unmet demand or a simplified transportation cost while respecting capacity constraints.

- Decision variable: amount of freight allocated to each corridor.
- Objective: minimize unmet demand or total modeled logistics cost.
- Constraints: available capacity, estimated demand and corridor limits.
- Output: a scenario showing how capacity could be distributed under the assumed conditions.

This is a planning framework rather than a recommendation for actual railway operations. Real operational optimization would require additional variables such as wagon availability, train paths, terminal capacity, transit times and operating costs.

15. Risks and Limitations

- The DGCIS publication is aggregate statistical data and does not provide real-time train or vehicle information.
- Freight quantity does not directly represent transportation cost, profitability or service quality.
- The published commodity coverage is based on selected commodities/commodity groups.
- Historical movement patterns cannot by themselves establish causal relationships.
- Forecasting results will be limited by the number and consistency of historical years available.
- Optimization results will depend heavily on assumptions about demand, capacity and cost.
- Data extracted from PDF tables must be validated carefully because formatting errors can affect analysis.

16. Expected Outcomes

- A cleaned and documented Indian rail-freight dataset for analysis.
- A KPI framework for monitoring freight movement.
- State-wise and commodity-wise visual analysis.
- An origin-destination flow matrix for identifying major corridors.
- A state segmentation model based on freight characteristics.
- A roadmap for future freight-demand forecasting.

- A conceptual capacity-allocation optimization model.
- A concise set of evidence-based logistics insights for decision-makers.

17. Conclusion

This project provides a practical starting point for applying data science to Indian logistics. The main strength of the approach is that it begins with a clear business question and measurable KPIs before moving to advanced techniques. The DGCIS 2024-25 rail statistics provide enough structure to study state-level movement, commodity concentration and inter-state flows.

The next phase would be to work with the detailed tables in Python, validate the published totals, create visualizations and test whether clustering or forecasting adds useful information. Optimization can then be introduced with clearly stated assumptions. This staged approach reduces the risk of applying machine learning without first understanding the underlying logistics data.

18. References

8. Directorate General of Commercial Intelligence & Statistics (DGCIS), Inter/Intra-State Movements/Flows of Goods by Rail, River and Air – 2024-25.
9. Directorate General of Commercial Intelligence & Statistics (DGCIS), Publications: Inland Trade Statistics.
10. Directorate General of Commercial Intelligence & Statistics (DGCIS), Inter/Intra-State Movements/Flows of Goods by Rail, River and Air – 2023-24.
11. Government of India, Press Information Bureau, Indian Railways freight-loading information for FY 2024-25.

Official source:

DGCIS Inland Trade Statistics: https://www.dgciskol.gov.in/pub_inland.aspx

2024-25 publication:

https://www.dgciskol.gov.in/writereaddata/Downloads/20260227120851IT%20REPORT_2024_25_Final.pdf

19. Submission Note

This report is intended as the Week 1 strategic-planning deliverable. It defines the problem, identifies the relevant KPIs, documents the public data source, explains the proposed analytical methods and provides a roadmap for the implementation phase. Numerical findings quoted in the report are taken from the official DGCIS 2024-25 publication; proposed model outputs are clearly identified as future analysis.